



Ileana Gomez Leal, PhD



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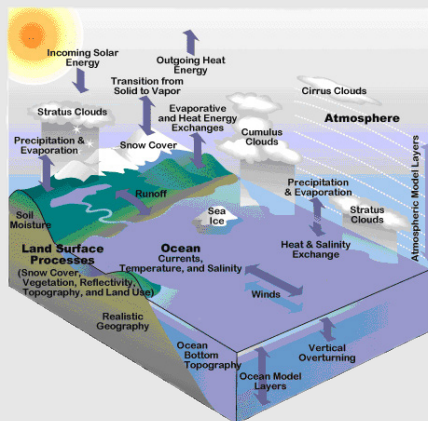
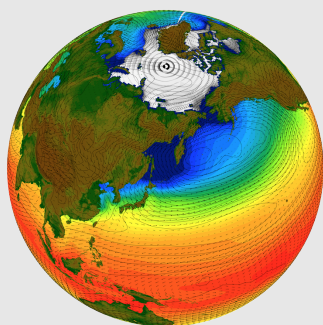
<https://www.researchgate.net/profile/Ileana-Gomez-Leal-2>



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Research interests

- Global climate models
- Atmospheric dynamics
- Habitability
- Planetary and exoplanetary atmospheres
- Paleoclimate and climate change
- Satellite observations, planetary imaging, 3D mapping.



Research Experience

- 2015-2026 Research Associate** Cornell University, Ithaca, USA
- <https://orcid.org/0000-0003-2254-5936>
 - Cornell Center for Astrophysics and Planetary Science (CCAPS)/ Earth and Atmospheric Sciences (EAS)
 - Global climate models (GCMs).
 - Habitability conditions of Earth-like planets and Super-Earths. Atmospheric dynamics of terrestrial planets. Solar Radiation Modification (SRM) studies to cool the Earth.
- 2014 Research Associate** MPIA, Heidelberg, Germany
- Variation of the thermal light curves of Earth-like planets and Super-Earths using GCMs.
 - Planet and Star Formation Department (PSF) at the Max-Planck Institut für Astronomie (MPIA):
http://mpia.de/Public/menu_q2.php?../PSF/contact.php
- 2009-2013 PhD. in Astrophysics** CNRS/Université de Bordeaux, France
- Spectrophotometry of the infrared emission of Earth-like planets. Advisor: Dr. Franck Selsis.
 - Laboratoire d'Astrophysique de Bordeaux. Université de Bordeaux:
https://tel.archives-ouvertes.fr/file/index/docid/962379/filename/GOMEZ_ILLEANA_2013_RED.pdf

Education

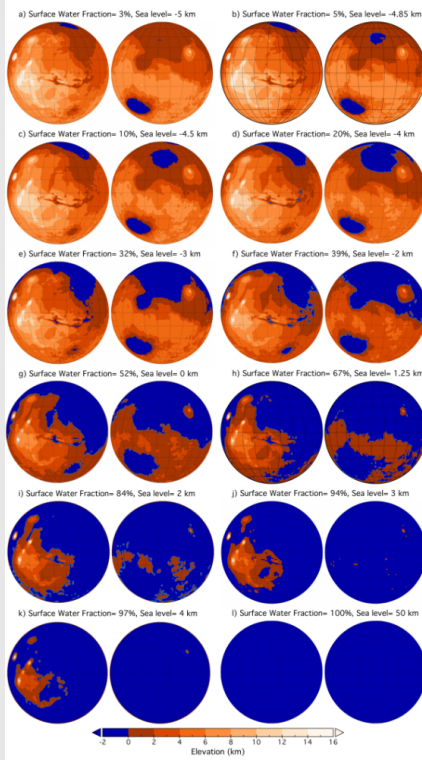
- 2013 Ph.D. in Astrophysics** CNRS/Université de Bordeaux, France
- Thesis topic: *Spectrophotometry of the infrared emission of Earth-like planets*. Advisor: Dr. Franck Selsis.
<https://tel.archives-ouvertes.fr/tel-00962379/>.
- 2009 Postgraduate Certificate in Education** Univ. Alfonso X El Sabio, Spain
- <https://www.uax.es/en/get-to-know-uax.html>
- 2009 M.Sc. in Astrophysics** Universidad de la Laguna, Spain
- Master thesis : *Photometric variability of the disk-integrated thermal emission of the Earth*. Advisor: Enric Pallé.
<https://www.ull.es/en/masters/astrophysics/>
- 2007 B.Sc. in Physics** Universidad Autónoma de Madrid, Spain
- Major in Theoretical Physics.
<https://www.uam.es/uam/grado>

Awards

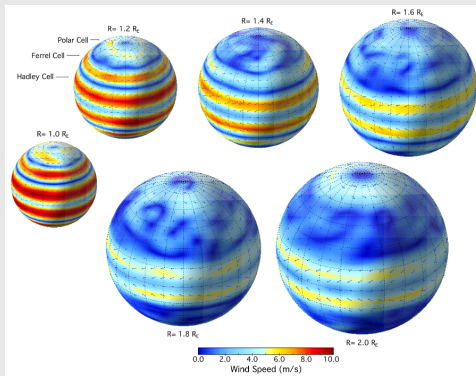
- 2009-2013 International Graduate Fellowship** IAC, Spain/CNRS, France
- International Fellowship (PhD. Program) from the Instituto Astrofisico de Canarias (IAC), Spain and the Centre National de la Recherche Scientifique (CNRS), France.
- 2007 ESAC Trainee** European Space Astronomy Centre (ESAC), Spain
- Study of the reddening correction from the 2175 Å interstellar dust absorption feature. Advisor: Dr. Carmen Morales-Duran.
<https://www.cosmos.esa.int/web/esac-trainees/trainees-2007>

Skills

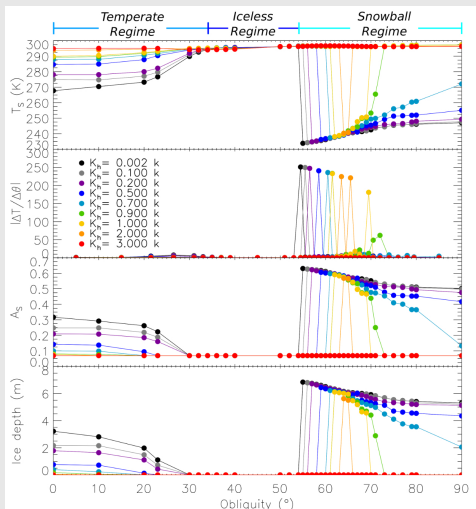
- Programming languages**
- Python, Fortran, IDL, Bash, Unix, LaTeX.
- Global Climate Models**
- LMDZ (Paris, France), PlaSim (Hamburg, Germany).
- Languages**
- Fluent in English, French, Italian, Spanish. Basic German.



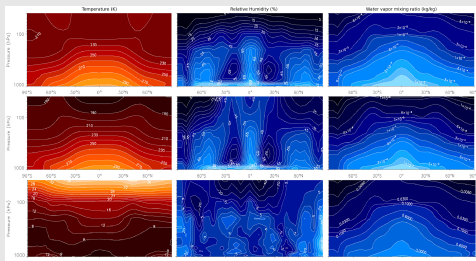
Habitability on Mars-size planets



Circulation of Super-Earth analogs



Climate sensibility to obliquity



Climate sensitivity to Ozone

Publications

- 2026 **Gomez-Leal, I.** *Water loss on Earth-like planets: evolution of irradiance and sea level.* In prep.
- 2026 **Gomez-Leal, I.,** Kaltenecker, L., Lucarini, V. & Lunkeit, F. *Climate sensitivity to obliquity and ocean heat transport on Earth-like planets.* In prep.
- 2026 **Gomez-Leal, I.** *Habitability of Mars-size planets: the importance of water content and surface albedo.* Submitted to Life, (ISSN 2075-1729; CODEN: LBSIB7). In review.
- 2026 **Gomez-Leal, I.,** Kaltenecker, L., & Lucarini V. *Cool Super Earths: a larger size reduces circulation and temperature* Submitted to ApJ. In review.
- 2020 Poggiali, V., Hayes, A. G., Mastrogiuseppe, M., Le Gall, A., Lalic, D., **Gomez-Leal, I.,** & Lunine, J. I. *The bathymetry of Moray sinus at Titan's Kraken Mare.* Journal of Geophysical Research: Planets, 125, e2020JE006558. <https://doi.org/10.1029/2020JE006558>
- 2019 **Gomez-Leal, I.,** Kaltenecker, L., Lucarini, V., & Lunkeit, F. 2019. *Climate sensitivity to ozone and its relevance on the habitability of Earth-like planets.* Icarus, 321, 608. <https://doi.org/10.1016/j.icarus.2018.11.019>
- 2018 **Gomez-Leal, I.,** Kaltenecker, L., Lucarini, V., & Lunkeit, F. 2018. *Climate Sensitivity to Carbon Dioxide and the Moist Greenhouse Threshold of Earth-like Planets under an Increasing Solar Forcing.* The Astrophysical Journal, 869, 2. <https://doi.org/10.3847/1538-4357/aaea5f>
- 2016 **Gomez-Leal, I.,** Codron, F., & Selsis, F. 2016. *Thermal light curves of Earth-like planets: 1. Varying surface and rotation on planets in a terrestrial orbit.* Icarus, 269, 98. <https://doi.org/10.1016/j.icarus.2015.12.050>
- 2015 Giovanna Tinetti, G., Drossart, P., Eccleston, P., and 352 coauthors, including **Gomez-Leal, I.** 2015. *The EChO science case.* Experimental Astronomy, 40, 329. <https://doi.org/10.1007/s10686-015-9484-8>
- 2012 **Gomez-Leal, I.,** Pallé, E., and Selsis, F. 2012. *Photometric Variability of the Disk-integrated Thermal Emission of the Earth.* The Astrophysical Journal, 752, 28. <https://doi.org/10.1088/0004-637X/752/1/28>

Selected contributions to Conferences

- 2020 ★ Talk.- P020-02-*The Importance of the Ocean Size on the Climate and the Water Loss of Mars-size planets.* AGU Fall Meeting, 1-17 December. <https://ui.adsabs.harvard.edu/abs/2020AGUFMP020...02G/abstract>
- 2017 ★ Talk.- Diversity of Planetary circulation regimes in our solar system and beyond. École de Physique, Les Houches, France.

Advisory Research and teaching experience

- 2015-2016 ★ Supervision of Research Experience for Undergraduates (REU) students and Graduate students at Cornell University.
- 2014 ★ Supervision of the Bachelor thesis of Nicolas Brod Bach and Anouk Ehreiser, students at Heidelberg University.

Science Vulgarization

- 2020-2021 Articles for the magazines: *L'Astronomie* (France), *Astronomia* (Spain), and *Cosmo* (Italy).